Web browser and type *http://localhost* on the address bar to check if XAMPP is working properly. If the XAMPP page loads correctly, proceed with the installation of Moodle. You can download the installation file from the official website of Moodle at *https://download.moodle.org/*. The latest version of Moodle 3.5 in different compression and archiving file formats is available for download at this location.

After downloading the installation file called *moodle-3.5.tgz*, extract and copy it into the directory */opt/lampp/htdocs* with admin privileges. Next, create a database for Moodle in phpMyAdmin, which is available with XAMPP. Open a Web browser and type *http://localhost/phpmyadmin* on the address bar to access phpMyAdmin. On the browser, click the option *New* to create a new database. Select the option *Create database* and name it 'moodle'. Using a Web browser, go to *http://localhost/moodle*. Open the Moodle configuration pages. While going through the Moodle installation process, name the database *moodle*—the name of the database we have created in phpMyAdmin. Setting MariaDB as the default database is beneficial if you are using XAMPP.

During the installation, you will be asked to provide certain paths and other options, yet on the whole, you will be able to complete the installation without much difficulty—just make sure that you have admin privileges. Following these steps will only enable you to install Moodle in your local system. To make your Moodle installation available publicly, you need to purchase a domain name and server space from a Web hosting company and configure the cPanel — a Web-based hosting control panel offered by many hosting providers to website owners, allowing them to manage their websites from a Web based interface. You can then upload and extract the Moodle folder in the cPanel to make it available publicly.

The other option is to obtain just a domain name and configure Moodle in Google or Amazon cloud services and link that server's IP address to your own domain name. Before discussing CodeRunner and Virtual Programming Lab (VPL), let's create a course in Moodle. Courses are the spaces that allow Moodle users to create and provide learning materials to their students. Moodle directly provides options to create new courses and the operation is simple. In this article, we have created just a single course called *Programming* to make our discussion easy to follow.

It is not possible to include a large number of images in a single article so we have prepared a separate document titled *Configuration.pdf* which is available for download at *opensourceforu.com/article_source_code/July18moodle.zip*. The document consists of a detailed set of screenshots involving the configuration of Moodle as well as the configuration and use of the two plugins, CodeRunner and VPL. Please download and go through this document if you get confused with any of the steps involved in the configuration of Moodle, CodeRunner or VPL.

Installing CodeRunner

Now it is time to install the first plugin called CodeRunner, which is going to help programming language instructors a lot. Unlike other normal plugins in Moodle, CodeRunner and VPL involve a two-step installation process. First, we have to install the actual plugin and then install sandbox servers to run programs to be tested by CodeRunner and VPL. Sandbox servers are needed to prevent system crashes when students experiment with code, either due to the unintentional use of bad code or malicious code. Either way, it is not at all safe to allow third party programs to be executed by your own local compiler. Sandbox servers allow the third party programs to use only a restricted set of resources of the host machine and thus protect the system from any harm.

Moodle allows the course managers to set questions, the answers for which can be one among several question types. Some of the more commonly used question types of Moodle include multiple choices, short answers, numerical, descriptions, essays, matching, etc. CodeRunner offers question types that allow programming course content managers to set programming questions in which the student's answer is code in some programming language. This code can then be automatically graded by running and testing it against test cases provided by the course manager. CodeRunner can be used to conduct programming tests in languages like C, C++, Java, Python, JavaScript, PHP, MATLAB, etc. To install the CodeRunner plugin, you have to log in to your Moodle server as the administrator. The plugin can be installed directly from the Moodle plugins directory. To do this, select the menu item *Dashboard>Site administration>Plugins>Install plugins*. We have installed the plugin CodeRunner from a zip file provided by the Moodle plugins directory. Figure 1 shows the CodeRunner plugin installation.

Setting up a sandbox server for CodeRunner

The sandbox server for CodeRunner is called Jobe server. There are two options available while deploying the Jobe server or any other sandbox server. You can use the address of the default Jobe server provided online or you can set up your own Jobe server on your local system. In this article, we have used the default Jobe server available online at *jobe2.cosc.canterbury.ac.nz* provided by the University of Canterbury, New Zealand

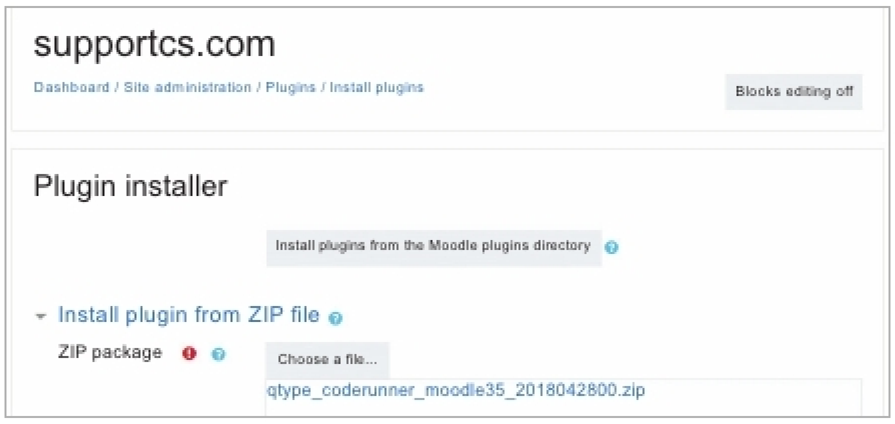

Figure 1: CodeRunner plugin installation